

Aniruddh Chandratre

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O-1A (current) · EB-1A approved

ML infrastructure engineer building distributed data platforms, GPU scheduling, and cluster-efficiency systems for FSD, Optimus, and Digital Optimus across large heterogeneous GPU fleets.

EXPERIENCE

Senior Engineer, ML Infrastructure — Tesla AI Feb 2024 – Present · Palo Alto, CA
[Distributed data platform](#) · [GPU scheduling](#) · [cluster efficiency](#)

- Lead Tesla AI's internal data processing platform for acquisition, curation, mining, auto-labeling, training, and evaluation; it runs **10B+ jobs** and **3,000+ compute-years/month** across **10** GPU clusters.
- Re-architected the platform core, distributed worker coordination, and memory model, increasing throughput from **940M to 15B+** data items/month.
- Built scheduling and live defragmentation for a heterogeneous GPU fleet, increasing usable compute capacity from **24% to 88%** and fleet-wide effective utilization to **98%**.
- Added end-to-end lineage across artifacts, datasets, training jobs, and models for multi-stage pipeline provenance.
- Built internet-scale acquisition and curation pipelines across Tesla AI programs.

Software Engineer, Automation Software — Tesla Jan 2023 – Jan 2024 · Fremont, CA
[Edge-to-cloud data infrastructure for the gigafactories](#)

- Designed and shipped an edge-to-cloud data broker processing **10B+ manufacturing data points/day** across Austin, Berlin, and Fremont.
- Developed an mmap-backed durable queue sustaining **5M writes/s** on NFS-backed Kubernetes volumes and a distributed cache processing **10M entries/s**, cutting storage **80%** across **50** deployments.
- Built and operated **20** high-availability services for factory insights.

Graduate Research Assistant — ASU, Cyber-Physical Systems Lab Jan 2021 – Dec 2022 · Tempe, AZ

- Researched formal verification under DARPA ARCOS with Lockheed Martin; co-authored PSY-TaLiRo and operated the Kubernetes cluster for high-volume simulation workloads.

Founding ML Engineer — Aftershoot Aug 2020 – Jan 2021 · Remote

- Built the image-ranking model at the core of AfterShoot's photo-culling product, now used by approximately **250K** photographers.

EDUCATION

M.S. Robotics & Autonomous Systems (AI) — Arizona State University · 4.0/4.0, Distinction Medal 2022
B.Tech Electronics & Telecommunications — NMIMS MPSTME, Mumbai · Dean's List 2020

RESEARCH, AWARDS & TOOLING

PUBLICATIONS [8 peer-reviewed papers](#) · [174 citations](#) — HSCC, EMSOFT, FMICS, ARCH. First author, HSCC 2023: *Stealthy Attacks Formalized as STL Formulas for Falsification of CPS Security*.

AWARDS 5× national hackathon winner in India — Smart India Hackathon 2019, Mastek Deep Blue (×2), IET Hack N Code, RAENG.

LANGUAGES Go, Python, C, C++, JavaScript/TypeScript, Java

SYSTEMS Slurm, Kubernetes, Helm, Docker, AWS, GCP, Prometheus, Grafana, Kafka, ClickHouse, InfluxDB, MQTT, SQL